## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Efficacy and Safety of Abdominal Paracentesis Drainage on Acute Pancreatitis Patients: A Systematic Review and Meta-Analysis |
| **AUTHORS** | Lu, Zongqing; Zhu, Xingxing; Hua, Tianfeng; Zhang, Jin; Xiao, Wenyan; Jia, Di; Yang, Min |

## VERSION 1 – REVIEW

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| **REVIEWER** | Popa , Cristian Constantin<br>Universitatea de Medicina si Farmacie Carol Davila, Surgery, Second Clinic of Surgery, Emergency University Hospital |
| **REVIEW RETURNED** | 28-Oct-2020 |

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| **GENERAL COMMENTS** | First of all, let me to congratulate you for writing this paper and for your entire work and efforts required to make this article!<br>The article addresses an important topic, relatively less studied, but of great relevance and clinical, social and economic importance.<br>All the characteristics necessary for the publication of a very good article have been successfully achieved.<br>Of course, the number of cohort studies you have studied in this article is limited, but this will certainly improve in the near future, given that it is a topic that has not been much studied.<br>I congratulate you once again for making this article. I hope it will be published by the management of BMJOpen journal as soon as possible and I wish you to continue your work on this choose topic and have as many professional, clinical and research success in your medical activity! |

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| **REVIEWER** | Voermans, Rogier<br>University of Amsterdam |
| **REVIEW RETURNED** | 19-Feb-2021 |

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| **GENERAL COMMENTS** | Hte authors present a well performed systematic review of an interesting concept. Although interesting some major issues need to be clarified. The most important flaw is that the their conclusion is only supported by the cohort studies. Not by the RCTs. PLease adapt the conclusion and discussion. |

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| **REVIEWER** | Chen, Lingxiao<br>The University of Sydney Institute of Bone and Joint Research |
| **REVIEW RETURNED** | 02-Apr-2021 |

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| **GENERAL COMMENTS** | Major concerns:<br>Please justify the reason to include observational studies considering your research question is about intervention. Please |

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| | keep in mind “We need less research, better research, and research done for the right reasons”.<br>Minor concerns:<br>1. Need a native speaker to improve language.<br>2. Some terms need clarification, such as organ failure, related infectious complications, and spending.<br>3. RoB 2 should be used to assess the risk of bias in RCTs. Please check BMJ. 2019 Aug 28;366:l4898.<br>4. Considering the risk of bias of all cohort studies, the current grade is high, which is unreasonable. NOS is an old tool. Currently, the Cochrane collaboration recommended ROBINS-I to assess the risk of bias in non-randomised studies of interventions. Please check BMJ 2016;355:i4919 and then use this tool to assess the risk of bias.<br>5. For continuous data, please use MD rather than SMD if possible.<br>6. Please do not choose the model based on the results from statistical tests.<br>7. Please set up the sensitivity analyses based on clinical knowledge and previous literature. The leave-one-out method is data-driven.<br>8. The rate of organ failure and the rate of related infectious complications might be affected by length of follow-up. Need sensitivity analysis or justify this issue. |

**VERSION 1 – AUTHOR RESPONSE**

Reviewer 1

1. First of all, let me to congratulate you for writing this paper and for your entire work and efforts required to make this article!
The article addresses an important topic, relatively less studied, but of great relevance and clinical, social and economic importance.

All the characteristics necessary for the publication of a very good article have been successfully achieved.

Of course, the number of cohort studies you have studied in this article is limited, but this will certainly improve in the near future, given that it is a topic that has not been much studied.

I congratulate you once again for making this article. I hope it will be published by the management of BMJOpen journal as soon as possible and I wish you to continue your work on this choose topic and have as many professional, clinical and research success in your medical activity!

Answer : It is our honor to achieve your approval. Acute pancreatitis is an acute and intense inflammatory disease process of the pancreas with an associated high mortality rate and financial burden in ICU. Abdominal paracentesis drainage as a novel treatment has been practiced in Chinese hospitals. The purpose of this study was to demonstrate the effectiveness of APD based on the currently available evidence and provide thoughts and inspiration for the follow-up high-quality research and clinical trials. We will keep an eye on this topic and expect that this aspect gets more attention. Finally, thanks again for your contribution to this study.

Reviewer 2

1. The authors present a well performed systematic review of an interesting concept. Although interesting some major issues need to be clarified. The most important flaw is that the their conclusion is only supported by the cohort studies. Not by the RCTs. Please adapt the conclusion and discussion.
Answer：First of all, thanks for your suggestions. In this study, a total of five cohort studies and three RCTs have been included. The effect of APD in-hospital duration and spending were both confirmed by cohort studies and RCTs. We fully agree with the major flaw you have raised. But it is worth noting that observational study is an important source of evidence when assessing new technology, especially if the amount of related RCTs is insufficient. The main objective of this study was to demonstrate the clinical, social and economic effectiveness of APD based on the currently available evidence, to provide thoughts and inspiration for the follow-up high-quality research and clinical trials. We recognize the instability of results may due to the risk of bias that existed in observational studies, and have made a special statement in the limitations. In this revision, we have adapted the conclusion and discussion and further emphasized that the pooled results depend largely on observational studies, which means that final pooled results must be interpreted very cautiously.

Reviewer 3

1. Please justify the reason to include observational studies considering your research question is about intervention. Please keep in mind “We need less research, better research, and research done for the right reasons”.
Answer：First of all, thanks for your constructive suggestions. Abdominal paracentesis drainage as a novel treatment for curing acute pancreatitis has been practiced in Chinese hospitals and showed a better therapeutic efficacy verified by several observational studies when compared with traditional methods. The observational study is an important source of evidence when assessing a new technology or lacking RCTs proving, although it has many inherent defects, such as a high risk of bias.

2. Need a native speaker to improve language.
Answer：We are so sorry for this point. In this revised version, we have improved language expression.

3. Some terms need clarification, such as organ failure, related infectious complications, and spending.
Answer：For this part, we want to explore the rate of organ failure, related infectious complications such as microbial infection, sepsis and bacteremia after treatment and total spending during hospitalization. We did not predefine the type of organ failure in order to comprehensively evaluate the effectiveness of APD. You can find this on page 7, line 13-14.

4. RoB 2 should be used to assess the risk of bias in RCTs. Please check BMJ. 2019 Aug 28;366:l4898.
Answer：We totally concur with your recommendation and made the corresponding modification. You can find it on Figure 2.

5. Considering the risk of bias of all cohort studies, the current grade is high, which is unreasonable. NOS is an old tool. Currently, the Cochrane collaboration recommended ROBINS-I to assess the risk of bias in non-randomised studies of interventions. Please check BMJ 2016;355:i4919 and then use this tool to assess the risk of bias.
Answer : First of all, thanks for this wonderful suggestion. When grading the risk of bias of included cohort studies, we have aware of this point. In this revision, we made the corresponding modification according to your recommendation and very discreetly downgraded two studies through discussion and consultation with relevant experts. You can find it in Table 2.

6. For continuous data, please use MD rather than SMD if possible.
Answer : Thanks for your suggestion. But given the inconsistencies in numerical measurements and units, we chose SMD carefully in the end.

7. Please do not choose the model based on the results from statistical tests.
Answer : Accoring to your suggestion, we have performed a discussion about this part. In the initial manuscript, we did not carefully consider the effect of small sample effects on the pooled results. The influence of large sample studies may be overamplified if the model is only based on heterogeneous results. So we used the random-effects model to merge each result. You can find this revision on page 8, lines 21-22.

8. Please set up the sensitivity analyses based on clinical knowledge and previous literature. The leave-one-out method is data-driven.
Answer : Thanks for your suggestion. We originally intended to conduct sensitivity analysis based on the etiology and canal diameter of APD. But the amount of data available did not support it. And there is not related previous literature available for reference. So we also eliminate low-quality studies to performed sensitivity analysis in this revision, and the results showed that these pooled results were still stable. We have described the limitations and expect future RCTs to address this area.

9. The rate of organ failure and the rate of related infectious complications might be affected by length of follow-up. Need sensitivity analysis or justify this issue.
Answer : We fully agree with your point of view. In the process of this revision, we have retrieved the following time of each study. But it is regrettable that we just get very rough data in the article. We then tried to get in touch with the author of the article but were ultimately unsettled. To show the importance of your point, we have made a special statement in the limitations, and expect future RCTs to address these issues.

**VERSION 2 – REVIEW**

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| **REVIEWER** | Voermans, Rogier<br>University of Amsterdam |
| **REVIEW RETURNED** | 27-May-2021 |

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| **GENERAL COMMENTS** | Major flaws discussed and conclusion adapted. |

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| **REVIEWER** | Chen, Lingxiao |

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| | The University of Sydney Institute of Bone and Joint Research |
| **REVIEW RETURNED** | 29-May-2021 |

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| **GENERAL COMMENTS** | Thanks for your hard work! The study quality is good now. But I still have two minor concerns.<br>1. For the data sources and searches section, it seems strange that the authors searched both OVID and EMBASE. In general, we could search Medline and EMBASE via OVID. OVID is a platform with many databases. Please clarify it.<br>2. I could still find quite a few grammatical errors. I understand that finding a native speaker might be difficult for a team with busy clinicians. May try a professional editorial service? |

## VERSION 2 – AUTHOR RESPONSE

Reviewer 2

1.Major flaws discussed and conclusion adapted.

Answer: Thanks for your great suggestions and contributions.

Reviewer 2

1. For the data sources and searches section, it seems strange that the authors searched both OVID and EMBASE. In general, we could search Medline and EMBASE via OVID. OVID is a platform with many databases. Please clarify it.
Answer : We are so sorry for this mistake. In this study, the OVID indicate the MEDLINE(OVID), and it has been modified.

2. I could still find quite a few grammatical errors. I understand that finding a native speaker might be difficult for a team with busy clinicians. May try a professional editorial service?

Answer : Sorry for the poor expression. In this revision, the manuscript has been improved by a professional editorial service.